

HW 2 Solutions

Part II:

1. (Full credit if question was fully answered)
2. Suppose right now it is raining, and I am drinking tea and reading a book, somewhere which is not my bedroom. Classify each of the following as being currently True, False, or Cannot be Determined. Explain your answers (a sentence for each is sufficient.)

- (a) If I am drinking tea right now, then I am reading a book.

Solution: $T \rightarrow T$ evaluates to **T**

- (b) If I am drinking tea right now, then I am in my bedroom.

Solution: $T \rightarrow F$ evaluates to **F**

- (c) If it is not raining right now, then I am drinking tea.

Solution: $F \rightarrow T$ evaluates to **T** (vacuously true)

- (d) If it is not raining right now, then I am riding a bicycle.

Solution: $F \rightarrow ?$ evaluates to **T** (vacuously true regardless of the value of "?")

- (e) If I am wearing purple right now, then I am drinking tea.

Solution: $? \rightarrow T$ evaluates to **T** (regardless of the value of "?")

- (f) If I am reading a book right now, then I am at the library.

Solution: $T \rightarrow ?$ evaluates to **Cannot Be Determined** (because the value of "?" makes a difference.)

3. Solve the following syllogisms by first rewriting each statement as a logical implication, then combining logical implications by transitivity, and finally writing the overall conclusion in English.

- (a)
- The only articles of food, that my doctor allows me, are not very rich.
 - Nothing that agrees with me is unsuitable for supper.
 - Wedding cake is always very rich.
 - My doctor allows me all articles of food that are suitable for supper.

Universe = "articles of food"; D = "doctor allows", R = "is very rich", A = "agrees with me", S = "is suitable for supper", W = "is wedding cake."

Solution

$$D \rightarrow \neg R$$

$$A \rightarrow S$$

$$W \rightarrow R$$

$$S \rightarrow D$$

We combine these as follows:

$$A \rightarrow S \rightarrow D \rightarrow \neg R \rightarrow \neg W$$

to conclude $A \rightarrow \neg W$, hence:

“Anything that agrees with me is not wedding cake.” Or one could use the contrapositive of all the above and conclude the more natural “Wedding cake does not agree with me.”

- (b)
- Nobody, who really appreciates Beethoven, fails to keep silence while the Moonlight-Sonata is being played;
 - Guinea-pigs are hopelessly ignorant of music;
 - No-one, who is hopelessly ignorant of music, ever keeps silence while the Moonlight-Sonata is being played.

Universe = “creatures”; G = guinea-pigs; I = hopelessly ignorant of music; S = keeping silence while the Moonlight-Sonata is being played; B = really appreciating Beethoven.

Solution

$B \rightarrow S$ [Note: this sentence might be more directly translated as $\neg(B \wedge \neg S)$, which is logically equivalent to $B \rightarrow S$.]

$$G \rightarrow I$$

$$I \rightarrow \neg S$$
 [from the more direct translation $\neg(I \wedge S)$.]

We combine these as follows:

$$G \rightarrow I \rightarrow \neg S \rightarrow \neg B$$

to conclude $G \rightarrow \neg B$, hence:

“Guinea-pigs definitely do not appreciate Beethoven.”

Part III, from *Scheinerman* do the following problems. “2nd{x.y}” refers to problem x.y in the 2nd edition and “3rd{x.y}” refers to problem x.y in the 3rd edition.

Please note that what we are most interested in is the clarity of your explanations, not just the correct answer.

Using **3rd** edition:

5.6

Solution

Let x, y be odd. There exist integers a, b such that $x = 2a + 1$ and $y = 2b + 1$ and so $xy = (2a + 1)(2b + 1) =$

$4ab + 2a + 2b + 1 = 2(2ab + a + b) + 1$. Since $2ab + a + b \in \mathbb{Z}$, then by definition xy is odd.

5.9

Solution

Suppose $a|b$ and $a|c$. By definition, there are integers x and y with $b = ax$ and $c = ay$. Therefore $b + c = ax + ay = a(x + y)$. Since $x + y \in \mathbb{Z}$, then by definition $a|(b + c)$.

5.15

Solution

($\Rightarrow$) Assume $0 \mid x$. By definition 3.2 there is an integer c such that $0 * c = x$. Therefore $x = 0$.
($\Leftarrow$) Assume $x = 0$. We want to show $0 \mid 0$. This is true by definition 3.2 because there is an integer c such that $0 * c = 0$. For example, $c = 1$ works.

6.1

Solution

There are many possible counterexamples. Let $a = 3$ and $b = -6$. Then $a|b$ but $a > b$.

6.2

Solution

Let $a = 10$ and $b = 0$. Then $a|b$ but $a > b$.

6.11

Solution

$x = 0$ is a counterexample; the only counterexample.

7.3

Solution

To prove: $(x \wedge y) \vee (x \wedge \neg y)$ is logically equivalent to x

Proof:

x	y	$\neg y$	$x \wedge y$	$x \wedge \neg y$	$(x \wedge y) \vee (x \wedge \neg y)$
T	T	F	T	F	T
T	F	T	F	T	T
F	T	F	F	F	F
F	F	T	F	F	F

Since the first column and last column contain identical truth values in each row, we have shown that $(x \wedge y) \vee (x \wedge \neg y)$ is logically equivalent to x .

■ *Note: For 7.3 the textbook directions do not specify using a truth table. It is also possible to prove this by applying the Boolean properties. We don't cover this method in this class but a student can get credit for doing it this way.*

7.9

Solution

$2^{10} = 1024$. Each variable has two possible values: True or False. Thus, there are 1024 total rows.

7.10c

Solution

To disprove a logical equivalence, we can create a truth table. If there exists a row in the truth table such that the two statements contain a different truth value, then the two statements cannot be logically equivalent.

To prove: $x \vee y$ is not logically equivalent to $(x \wedge \neg y) \vee ((\neg x) \wedge y)$

Proof:

x	y	$\neg x$	$\neg y$	$x \wedge \neg y$	$\neg x \wedge y$	$(x \wedge \neg y) \vee ((\neg x) \wedge y)$	$x \vee y$
T	T	F	F	F	F	F	T
T	F	F	T	T	F	T	T
F	T	T	F	F	T	T	T
F	F	T	T	F	F	F	F

Since the last two columns do not contain identical truth values in the first row, we have shown that $x \vee y$ is not logically equivalent to $(x \wedge \neg y) \vee ((\neg x) \wedge y)$.

Note: this latter expression is known as “exclusive or”.

■

7.11b

Solution

To prove: $(x \wedge (x \rightarrow y)) \rightarrow y$ is a tautology

Proof:

x	y	$x \rightarrow y$	$x \wedge (x \rightarrow y)$	$(x \wedge (x \rightarrow y)) \rightarrow y$
T	T	T	T	T
T	F	F	F	T
F	T	T	F	T
F	F	T	F	T

Since the last column contains only TRUE, $(x \wedge (x \rightarrow y)) \rightarrow y$ is a tautology. ■

7.15

Solution

Finding an expression: In class we discussed that $x \leftrightarrow y$ is logically equivalent with $(x \rightarrow y) \wedge (y \rightarrow x)$. However, we also know $x \rightarrow y = (\neg x) \vee y$ (given in the exercise prompt). Thus, an equivalent expression should be

$$x \leftrightarrow y = (x \rightarrow y) \wedge (y \rightarrow x) = ((\neg x) \vee y) \wedge ((\neg y) \vee x).$$

We will prove this using a truth table.

Claim: $x \leftrightarrow y = ((\neg x) \vee y) \wedge ((\neg y) \vee x)$

Proof:

x	y	$\neg x$	$\neg y$	$x \leftrightarrow y$	$(\neg x) \vee y$	$(\neg y) \vee x$	$((\neg x) \vee y) \wedge ((\neg y) \vee x)$
T	T	F	F	T	T	T	T
T	F	F	T	F	F	T	F
F	T	T	F	F	T	F	F
F	F	T	T	T	T	T	T

Since the fifth column and last column contain identical truth values in each row, we have shown that $x \leftrightarrow y$ is logically equivalent to $((\neg x) \vee y) \wedge ((\neg y) \vee x)$. ■

7.17

Solution

For every x, y combination in the chart, $x * y$ can either be TRUE or FALSE (2 possible values). Since there are 4 combinations of x, y , and each combination can take on 2 values, we have $2 \cdot 2 \cdot 2 \cdot 2 = 2^4 = 16$ ways to complete the chart. One possible way to complete the chart is the following:

x	y	$x * y$
TRUE	TRUE	TRUE
TRUE	FALSE	TRUE
FALSE	TRUE	TRUE
FALSE	FALSE	TRUE

Similarly, you can draw 15 more truth tables showing different patterns for the $x * y$ output.

7.18

Solution (credit to Ava Masse)

x	y	tautology $\top$ $x \vee \neg x$	or $\vee$ $x \vee y$	y implies x $y \rightarrow x$ $x \vee \neg y$	x implies y $x \rightarrow y$ $y \vee \neg x$	nand $\neg(x \wedge y)$	xor $\underline{\vee}$ $(x \vee y) \wedge \neg(x \wedge y)$	iff/nxor $x \Leftrightarrow y$ $\neg((x \vee y) \wedge \neg(x \wedge y))$
T	T	T	T	T	T	F	F	T
T	F	T	T	T	F	T	T	F
F	T	T	T	F	T	T	T	F
F	F	T	F	T	T	T	F	T

$\neg y$	$\neg x$	and $\wedge$ $x \wedge y$	x not implies y $\neg(y \vee \neg x)$	y not implies x $\neg(x \vee \neg y)$	nor $\neg(x \vee y)$	contradiction $\perp$ $x \wedge \neg x$
F	F	T	F	F	F	F
T	F	F	T	F	F	F
F	T	F	F	T	F	F
T	T	F	F	F	T	F